



Presentation

the open source programming tools available for it."

So when you choose components for your project, remember Denmead's thumb rule: "Be aware of what you need, and what's available to fit that need. You may not need the most expensive item for your project, and if you do your research, you can save some cost."

Acquiring the components

When it comes to acquiring the components, you have the choice of either buying them individually or going for a kit. You could also opt to buy on foot or order online (suggestions in box).

Acquiring individual components in India can be quite tough, unless you know where to go. The cost of components can also be high, because most of these are imported.

"Very few components get manufactured in India, so barring some passive components, all others are imported," says Neena Chopra, director, Kitsnspares.com, one of India's leading electronics DIY kit sellers.

In fact, cost is one of the major stumbling blocks felt by Indian Arduino users too. Priya Kuber, managing director of Arduino India, says, "The majority of Indian users, students and prototypers have complained about the high cost. Arduino as a project was made to cost as much as a textbook in Europe but when converted to Indian rupees it costs slightly higher to the Indian masses."

However, Kuber shares a good news too: "Arduino has decided to

bring out official 'made in India' boards soon!"

Overall, it appears that in India, it is easier and safer to buy kits. There are kits available for beginners as well as experts. The range includes specific projects as well as general-purpose ones, based on specific boards. Kits'n'Spares has DIY kits based on GSM, RF,

RFID, VLSI, etc.

Asimov Robotics also specialises in DIY kits, with offerings ranging from robot manipulators and humanoid platforms to animatronics platforms and virtual reality interfaces.

A third dimension—make it yourself

While we wonder whether to buy on foot or order online, several lucky individuals in the West are hardly bothered about the components, all thanks to 3D printing!

"With the ability to manufacture their own products, DIY manufacturers can create adventurous, unique products that were previously done only by manual methods. Clearly, the distance between the individuals and the manufacturers is gradually eliminated, by increasing the availability and affordability of 3D printers. In fact, the cost of 3D printers has dropped significantly in the last few years. Now almost anyone can own a 3D printer—MakerBot Replicator 2 with about \$2200—to create a 3D object in the real world," says Bajaj.

The viability of 3D printing has emerged in tandem with the advancement and democratisation of 3D design or CAD software, which allows users to easily create a digital 3D model of an object and optimise the design before

anything has been built. The combination of these two technologies—3D design software and 3D printers—means that it is easier for individuals to take an idea for a product and turn it into a physical object.

Although 3D printers are still quite expensive in India, you might want to co-invest in them with a DIY community. Plus, Bajaj feels, the prices are likely to go down as the market grows.

Which board?

When deciding on the components and tools needed, you are most likely to face the question "which board to use for my project." Kashinath says, "Among the group, members own several types of boards—Arduino, Raspberry Pi, Beagleboard, Stellaris, MSP430, etc. We basically use whatever serves our purpose for a particular project. However, choosing the right board (and hardware in general) is a recurring challenge for us and we are trying to develop a knowledge base around it. Not sure how easy it would be to keep it updated and accurate, given the rate at which new and improved board variants are coming out."

"There are many different boards, each having strengths in different areas. There are significant differences between what an Arduino is good for, since it is a microcontroller, versus what a Raspberry Pi is good for, as it is a minicomputer that runs an operating system. Because of these differences, we tend to publish more Arduino/microcontroller projects as they are used more commonly as part of a variety of purposes," says Denmead.

Sarafan appears to be an Arduino fan too. "I am a fan of Arduino because it is the most user-friendly and approachable. It has the same limitations as something like a Raspberry Pi, which also makes it more approachable to a more generalised audience. It is by no means the most powerful piece of hardware, but it is the most powerful in enabling absolute beginners to hit the ground running," he says.

Resources for components

- AliExpress
- Element14
- ExploreLabs
- Futurlec
- Kitsnspares.com
- SimpleLabs